FORM 2

THE PATENTS ACT, 1970

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The Patent Rules, 2003

COMPLETE SPECIFICATION

(See sections 10 & rule 13)

1. TITLE OF THE INVENTION

A PLANAR-COIL VARIABLE RELUCTANCE BASED ANGLE SENSOR SYSTEM WITH PWM OUTPUT

2. APPLICANT (S)

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3. PREAMBLE TO THE DESCRIPTION

COMPLETE SPECIFICATION

The following specification particularly describes the invention and the manner in which it is to be performed.

TECHNICAL FIELD

[0001] The present disclosure relates generally to detection systems. More particularly, the present disclosure relates to a planar-coil variable reluctance-based sensor system and method for detecting an angle of rotation of an object.

BACKGROUND

[0002] Absolute angle estimation plays a key role in automotive, industrial, robotic, aerospace, and biomedical applications. In many of these applications, performance metrics of angle sensors such as accuracy, precision, repeatability, and insensitivity to misalignments decide robustness of systems. Design metrics of the sensors such as affordability, minimal space complexity, operating temperature range, insensitivity to harsh and hostile environments such as presence of oil, dust, and humidity, are essential to integrate the angle sensor into any desired system. The angle sensors are widely used for positioning of rotary actuators, spindle tools in CNC machines, precise joint positioning and end-effector control in robotics, steering wheel rotation detection and throttle positioning in the automotive sector, and for tracking the position of control surfaces like rudders in aircraft.

[0003] Existing angle sensor technologies, such as resistive, capacitive, magnetic, optical, and planar coil-based sensors have distinct limitations that can impact their suitability for various applications. Resistive sensors are prone to mechanical wear due to physical contact, leading to reduced lifespan and potential inaccuracies. Capacitive sensors, while offering non-contact measurement, are highly sensitive to environmental factors such as humidity and temperature, which can cause drift and require frequent recalibration. Magnetic sensors, despite their robustness, can be affected by external magnetic fields and temperature variations, necessitating careful shielding and compensation mechanisms. Optical sensors, known for high precision, require a clear line of sight and can be disrupted by dust, fog, or ambient light conditions, limiting their applicability in certain environments. Further, planar coil-based sensors, though compact and cost-effective, are susceptible to misalignment, lift-off, and electromagnetic interference, which can degrade performance and accuracy. However, the existing sensor systems require complex signal processing and exhibit non-linear magnetic field variations with rotor movement, leading to delays unsuitable for high-speed applications.

[0004] Therefore, there is, a need for a high-speed, robust magnetic sensing system unaffected by noise, temperature, or rotor misalignments, which is ideal for industrial applications.

OBJECTS OF THE PRESENT DISCLOSURE

[0005] A general object of the present disclosure is to overcome the above-mentioned issues, and provide a system and a method for detecting angle of rotation of an object.

[0006] Another object of the present disclosure is to provide a planar coil based sensor for detecting the angle of detection of an object.

[0007] Yet another object of the present disclosure is to provide an interfacing module for determining the angle of rotation of the object from inductance variation of planar coils and for providing a PWM output corresponding to the angle of rotation of an object.

SUMMARY

[0008] The present disclosure relates generally to a sensor system. In particular, the present disclosure relates to a planar coil-based sensor system and method for precise measurement of rotational angle of an object. The planar coil-based sensor system may provide a full-circle (0◦-360◦) angle measurement using a variable reluctance principle. The system may include a sensor and an interfacing module. The sensor may incorporate two pairs of planar inductor coils wrapped around a shaft to form a cylindrical structure, with a thin ferromagnetic material attached to the shaft. As the shaft rotates, the inductance of each coil may change, enabling precise angle measurement. The system may generate a Pulse-Width Modulation (PWM) output, where on-time of the PWM output is directly proportional to the angle of rotation. The system offers a high-speed linear conversion of detected angle to the PWM output, and the system is immune to misalignment of the shaft and variations in permeability of the ferromagnetic strip.

[0009] In an aspect, the present disclosure relates to a system for detecting an angle of rotation of an object. The system may include a sensor configured with a shaft attached to the object. The shaft may be surrounded by one or more stationary coils and a ferromagnetic strip positioned on the shaft. An inductance of the stationary coils may varies in accordance with a movement of the ferromagnetic strip. Further, the system may include an interfacing module that may be electrically connected to the sensor. The interfacing module may be configured to convert inductance variations of the stationary coils to a Pulse Width Modulation (PWM) output to detect the angle of rotation of an object.

[0010] In another aspect, the present disclosure relates to a method for detecting an angle of rotation of an object. The method may include detecting, by a system, an inductance variation in one or more stationary coils based on the angle of rotation of an object. The method may include converting, by the system, the inductance variation in one or more stationary coils to voltage levels. Further, the method may include determining, by the system, a phase difference between the voltage levels associated with inductance variation in one or more stationary coils. Furthermore, the method may include converting, by the system, the phase difference between the voltage levels to a Pulse Width Modulation (PWM) output to detect the angle of rotation of an object.

[0011] In an embodiment, the stationary coils may include at least two pairs of square shaped planar inductor coils positioned around the shaft to form a hollow cylindrical structure.

[0012] In an embodiment, the two pairs of square shaped planar inductor coils may include a first pair of planar inductor coils that is positioned around the shaft comprised in the system and a second pair of planar inductor coils disposed surrounding a first surface of the first pair of planar inductor coils.

[0013] In an embodiment, each of the first pair and second pair of planar inductor coils may be bent over the shaft to form the hollow cylindrical structure.

[0014] In an embodiment, the first pair of planar inductor coils may be arranged such that one planar coil of the first pair covers an angle of 0^0 to 180^0 and the second planar coil of the first pair covers an angle of 180° to 360° of the hollow cylindrical structure.

[0015] In an embodiment, the second pair of planar inductor coils may be arranged such that one planar coil of the second pair covers an angle of 90^0 to 270^0 and the second planar coil of the second pair covers an angle of 270^0 to 90^0 of the hollow cylindrical structure.

[0016] In an embodiment, a predetermined air gap may be provided between the shaft and the one or more stationary coils.

[0017] In an embodiment, the interfacing module my include an inductance to voltage converter, a phase difference determination circuit and a phase to PWM converter.

[0018] In an embodiment, the inductance to voltage converter may be configured to convert the inductance variation in the stationary coils to corresponding voltage levels.

[0019] In an embodiment, the phase difference determination circuit may be configured to determine phase of the voltage levels.

[0020] In an embodiment, the phase to PWM converter may be configured to convert phase difference between the voltage levels associated with the inductance variation in the one or more stationary coils to the PWM output.

BRIEF DESCRIPTION OF DRAWINGS

[0021] The accompanying drawings, which are incorporated herein, and constitute a part of this invention, illustrate exemplary embodiments of the disclosed methods and systems in which like reference numerals refer to the same parts throughout the different drawings. Components in the drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the present invention. Some drawings may indicate the components using block diagrams and may not represent the internal circuitry of each component. It will be appreciated by those skilled in the art that the invention of such drawings includes the invention of electrical components, electronic components, or circuitry commonly used to implement such components.

[0022] The diagrams are for illustration only, which thus is not a limitation of the present disclosure, and wherein:

[0023] FIG. 1 illustrates an example block diagram of a system for detecting an angle of rotation of an object, in accordance with an embodiment of the present disclosure.

[0024] FIG. 2A illustrates a plan view of a sensor for detecting the angle of rotation of an object, in accordance with an embodiment of the present disclosure.

[0025] FIG. 2B illustrates a representation of a stationary coil on a flat surface and when it is bent over a semicircular arc, in accordance with an embodiment of the present disclosure.

[0026] FIG. 2C illustrates a flattened view of the relative position of stationary coils, in accordance with an embodiment of the present disclosure.

[0027] FIG. 2D illustrates an electrical circuit equivalent of the sensor, in accordance with an embodiment of present disclosure.

[0028] FIG. 2E illustrates a top view of the sensor, in accordance with an embodiment of the present disclosure.

[0029] FIG. 3A- 3E illustrates waveforms of the sensor, in accordance with an embodiment of the present disclosure.

[0030] FIG. 4A illustrates an example circuit diagram of an interfacing module of the system for detecting the angle of rotation of an object, in accordance with an embodiment of the present disclosure

[0031] FIG. 4B illustrates voltage waveforms of the interfacing module, in accordance with an embodiment of the present disclosure.

[0032] FIG. 5 illustrates an example flow chart for implementing a method for detecting an angle of rotation of an object, in accordance with an embodiment of the present disclosure.

[0033] FIG. 6A- 6C illustrates response waveforms of the system in an example implementation, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0034] In the following description, for the purposes of explanation, various specific details are set forth in order to provide a thorough understanding of embodiments of the present disclosure. It will be apparent, however, that embodiments of the present disclosure may be practiced without these specific details. Several features described hereafter can each be used independently of one another or with any combination of other features. An individual feature may not address all of the problems discussed above or might address only some of the problems discussed above. Some of the problems discussed above might not be fully addressed by any of the features described herein.

[0035] The ensuing description provides exemplary embodiments only, and is not intended to limit the scope, applicability, or configuration of the disclosure. Rather, the ensuing description of the exemplary embodiments will provide those skilled in the art with an enabling description for implementing an exemplary embodiment. It should be understood that various changes may be made in the function and arrangement of elements without departing from the scope of the disclosure as set forth.

[0036] Specific details are given in the following description to provide a thorough understanding of the embodiments. However, it will be understood by one of ordinary skill in the art that the embodiments may be practiced without these specific details. For example, circuits, systems, networks, processes, and other components may be shown as components in a block diagram form in order not to obscure the embodiments in unnecessary detail. In other instances, well-known circuits, processes, algorithms, structures, and techniques may be shown without unnecessary detail in order to avoid obscuring the embodiments.

[0037] Also, it is noted that individual embodiments may be described as a process which is depicted as a flowchart, a flow diagram, a data flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed but could have additional steps not included in a figure. A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, etc. When a process corresponds to a function, its termination can correspond to a return of the function to the calling function or the main function.

[0038] The word “exemplary” and/or “demonstrative” is used herein to mean serving as an example, instance, or illustration. For the avoidance of doubt, the subject matter disclosed herein is not limited by such examples. In addition, any aspect or design described herein as “exemplary” and/or “demonstrative” is not necessarily to be construed as preferred or advantageous over other aspects or designs, nor is it meant to preclude equivalent exemplary structures and techniques known to those of ordinary skill in the art. Furthermore, to the extent that the terms “includes,” “has,” “contains,” and other similar words are used in either the detailed description or the claims, such terms are intended to be inclusive in a manner similar to the term “comprising” as an open transition word without precluding any additional or other elements.

[0039] Reference throughout this specification to “one embodiment” or “an embodiment” or “an instance” or “one instance” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present disclosure. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

[0040] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0041] The description of terms and features related to the present disclosure shall be clear from the embodiments that are illustrated and described; however, the present disclosure is not limited to these embodiments only. Numerous modifications, changes, variations, substitutions, and equivalents of the embodiments are possible within the scope of the present disclosure. Additionally, the present disclosure can include other embodiments that are within the scope of the claims but are not described in detail with respect to the following description.

[0042] The present disclosure relates to a system and method for precise rotational angle measurement of an object. The system may employ variable reluctance principle to achieve full-circle (0°–360°) angle measurement. The system may include a sensor and an interfacing module. The sensor may include two pairs of planar inductor coils wrapped around a shaft, forming a cylindrical structure, with a thin ferromagnetic material attached to the shaft. As the shaft rotates, inductance of each coil varies, enabling accurate angle measurement. The interfacing module of the system may generate a Pulse-Width Modulation (PWM) output, where on-time of the PWM signal may be directly proportional to the angle of rotation of an object. The system may provide high-speed linear conversion of the detected angle to the PWM output.

[0043] In an aspect, the present disclosure relates to a system for detecting an angle of rotation of an object. The system may include a sensor configured with a shaft attached to an object. The shaft may be surrounded by one or more stationary coils and a ferromagnetic strip positioned on the shaft. An inductance of the one or more stationary coils may varies in accordance with a movement of the ferromagnetic strip. Further, the system may include an interfacing module that is electrically connected to the sensor. The interfacing module may be configured to convert inductance variations of the one or more stationary coils to a Pulse Width Modulation (PWM) output to detect the angle of rotation of an object.

[0044] In another aspect, the present disclosure relates to a method for detecting an angle of rotation of an object. The method may include detecting, by a system, an inductance variation in one or more stationary coils based on the angle of rotation of an object. The method may include converting, by the system, the inductance variation in one or more stationary coils to voltage levels. Further, the method may include determining, by the system, a phase difference between the voltage levels associated with inductance variation in one or more stationary coils. Furthermore, the method may include converting, by the system, the phase difference between the voltage levels to a Pulse Width Modulation (PWM) output to detect the angle of rotation of an object.

[0045] Various embodiments of the present disclosure will be explained in detail with reference to FIGs. 1-6.

[0046] Accurate measurement of rotational angles is fundamental across various applications, including automotive, industrial automation, robotics, aerospace, and biomedical systems. The present disclosure provides angle measurements by detecting changes in inductance corresponding to a ferromagnetic material movement with respect to the angle of rotation of an object. The present disclosure provides accurate angle measurements in harsh environments, immunity to misalignments, and minimal sensitivity to external magnetic interference. Additionally, the proposed planar coil-based sensor systems can be fabricated using cost-effective and compact methods, making them suitable for integration into a wide range of systems.

[0047] FIG. 1 illustrates an example block diagram of a system (102) for detecting an angle of rotation of an object, in accordance with an embodiment of the present disclosure.

[0048] Referring to FIG. 1, in an embodiment, the system (102) may include a sensor (104) and an interfacing module (106). The sensor (102) may be configured with a shaft (204) attached to an object whose rotational angle to be measured. The shaft (204) may be surrounded by stationary coils (202), and a ferromagnetic strip (206) may be positioned on the shaft (204). As the shaft (204) rotates, the movement of the ferromagnetic strip (206) may induce inductance variations of the stationary coils (202). The inductance variations may be converted into a Pulse Width

Modulation (PWM) output that corresponds to the angle of rotation of an object by the interfacing module (106) that may be electrically connected to the sensor (104).

[0049] FIG. 2A illustrates a plan view of a sensor for detecting the angle of rotation of an object, in accordance with an embodiment of the present disclosure.

[0050] FIG. 2B illustrates a representation of a stationary coil on a flat surface and when it is bent over a semicircular arc, in accordance with an embodiment of the present disclosure.

[0051] FIG. 2C illustrates a flattened view of the relative position of stationary coils, in accordance with an embodiment of the present disclosure.

[0052] FIG. 2D illustrates an electrical circuit equivalent of the sensor, in accordance with an embodiment of present disclosure.

[0053] FIG. 2E illustrates a top view of the sensor, in accordance with an embodiment of the present disclosure.

[0054] FIGs. 3A- 3E illustrates waveforms of the sensor (104), in accordance with an embodiment of the present disclosure.

[0055] Referring to FIGs. 2A to 2E and 3A to 3E, in an embodiment, the sensor (104) may include the shaft (204) (a movable part) and the stationary coils (202). The shaft (204) may be a cylindrical shaft with the ferromagnetic strip (206) affixed parallel to its axis. The stationary coils (202) may be square shaped planar coils. The stationary coils (202) may form a hollow cylindrical structure, around which four square-shaped planar coils (say X_1, X_2 X_3 and X_4) may be wrapped. Each of the square-shaped planar coil spans 180^0 of the full circle as shown in FIG. 2B. The stationary coils (202) X_1 and X_2 may be arranged such that X_1 covers 0^0 to 180^0 range and X_2 covers 180◦ to 360◦, as shown in FIG. 2E. Similarly, X_3 spans 90^0 to 270^0, while X_4 extends from 270◦ to 90◦. Thus, the stationary coil pairs $X_1 - X_2$ and $X_3 - X_4$ exhibit a 90^0 circumferential phase shift. The flattened view of the stationary coil (202) arrangement shown in FIG. 2C. The electrical circuit equivalent of the sensor shown in FIG. 2D, where $L_{X1} - L_{X4}$ corresponds to the inductances of $X_1 - X_4$ the stationary coils (202) respectively. The circular movement of the ferromagnetic material may varies the inductances ($L_{X1} - L_{X4}$) experienced by the stationary coils (202), that may be used in measuring the angle.

[0056] In an embodiment, a cylindrical structure may be provided to the sensor for supporting the stationary coils. The cylindrical structure may be made of materials such as polylactic acid monomer (PLA).

[0057] As shown in FIG. 2D, each of the stationary coils (202) X_1, X_2, X_3 and X_4 functions as a variable inductor with inductances L_{X1}, L_{X2}, L_{X3} and L_{X4}, respectively. In the absence of the ferromagnetic strip (206), each inductor maintains a nominal inductance of L_0. As the ferromagnetic strip (206) moves across each of the stationary coil (202) from $-x_m$ to $+x_m$, the inductance (L_X) of the stationary coils (202) varies in a sinusoidal manner. This variation is mathematically represented as,

$$L_X = L_0 + A\sin x \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (1)$$

Where, A is the maximum inductance experienced at x = 0. This variation in inductance of the stationary coil (202) as the ferromagnetic strip (206) moves across it from $-x_m$ to $+x_m$ is depicted in FIG. 3A. In the sensor (104), each of the stationary coil (202), extending from $-x_m$ to $+x_m$, may be wrapped around the hollow cylindrical structure. The stationary coils (202) may be wrapped around in such a way that each of the stationary coils (202) covers a semicircular perimeter of the hollow cylindrical structure, with a 90^0 offset between adjacent coils. When the shaft (204) rotates from 0^0 to 180^0, only the inductance of L_{X1} changes, while L_{X2} remains unaffected as the ferromagnetic strip (206) is outside its vicinity. Similarly, as the shaft (204) rotates from 180^0 to 360^0, L_{X2} undergoes an inductance change, while L_{X1} stays constant, as shown in FIG. 3B. Similarly, the inductances L_{X3} and L_{X4} vary with the shaft (204) position, exhibiting a 90^0 phase shift as shown in FIG. 3C. The inductance variation may cause responses relative to a reference point to follow a cosine pattern. The differential inductances L_{X1} - L_{X2} and L_{X3} - L_{X4} may follow pure sine and cosine waveforms, as illustrated in FIG. 3C. Referring to FIG. 3C, for each point along the angle axis, from 0^0 to 360^0, the sensor (104) may detect distinct combinations of L_{X1} - L_{X2} and L_{X3} - L_{X4}. Further, the angle of rotation may be determined through the interfacing module (106).

[0058] FIG. 4A illustrates an example block diagram (400A) of the interfacing module (106) of the system (102) for detecting the angle of rotation of an object, in accordance with an embodiment of the present disclosure.

[0059] FIG. 4B illustrates voltage waveforms of the interfacing module (106), in accordance with an embodiment of the present disclosure.

[0060] Referring to FIG. 4A and 4B, in an embodiment, the interfacing module (106) may include an inductance to voltage converter (402), a phase difference determination circuit (404) and a phase to PWM converter (406). The inductance to voltage converter (402) may include op-amps OA_1, OA_2, OA_3 and OA_4, and may be coupled to the stationary coils (202), L_{X1}, L_{X2}, L_{X3} and L_{X4}, of the sensor (104). Further, the inductance to voltage converter (402) may include a resistor R_S and sinusoidal voltage sources V_{S1} (= $V_M \sin(\omega t)$) and V_{S2} (= $V_M \cos(\omega t)$). A current of magnitude V_{S1}/R_S flows through the sensor (104) and the stationary coils (202) L_{X1} and L_{X2} may generate voltages V_{X1} and V_{X2} given as

$$V_{X1} = -j\omega L_{X1} V_M \sin(\omega t)/R_S \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(2)$$

$$\text{and } V_{X2} = -j\omega L_{X2} V_M \sin(\omega t)/R_S. \qquad (3)$$

Similarly, a constant current of V_{S2}/R_S may be injected into the stationary coils (202) L_{X3} and L_{X4} and may generate voltages, respectively, given as

$$V_{X3} = -j\omega L_{X3} V_M \cos(\omega t)/R_S \ (4)$$

$$\text{and } V_{X4} = -j\omega L_{X2} V_M \sin(\omega t)/RS, \qquad (5)$$

The output voltages V_{X1}, V_{X2}, V_{X3} and V_{X4} may be injected into the phase difference determination circuit (404), such as formed using an instrumentation amplifiers INA_1, INA_2 and INA_3, as shown in FIG. 4A. The output voltages of INA_1 and INA_2 is given as,

$$V_{O1} = j\omega(L_{X1} - L_{X2}) V_M \sin(\omega t) / R_S$$

$$= j\omega A V_M \sin\theta \sin(\omega t) / R_S \qquad (6)$$

$$V_{O2} = j\omega(L_{X3} - L_{X4}) V_M \cos(\omega t) / R_S$$

$$= j\omega A V_M \cos\theta \cos(\omega t) / R_S \qquad (7)$$

The voltage V_{O1} and V_{O2}, given in (6) and (7), are subtracted using INA_3, generating a voltage V_{O3} given as,

$$V_{O3} = j\omega AV_M (\sin\theta\sin(\omega t) - \cos\theta\cos(\omega t)) / R_S$$

$$= j\omega AV_M\cos(\omega t + \theta) / R_S \quad (8)$$

From equation (8), a measured angle (θ) may obtain that is directly proportional to phase of the output signal V_{O3}. Hence, by measuring the phase of V_{O3} relative to V_{S2}, the angle of rotation of an object may be determined. The phase difference between V_{O3} and V_{S2} may be measured by the phase to PWM converter (406) of the interfacing module (106). The phase to PWM converter (406) may detect the phase difference between V_{O3} and V_{S2} as on-time of a PWM wave. The phase to PWM converter (406) may include comparators OC_1 and OC_2, two D-flip-flops DFF_1 and DFF_2, and an XOR gate G_1. The voltage signals V_{O3} and V_{S2} are fed to the comparators to generate square pulse signals V_{OC1} and V_{OC2} respectively, as shown in FIG. 4B. The square pulse signals V_{OC1} and V_{OC2} may fed to DFF_1 and DFF_2 as clock signals. The DFF_1 and DFF_2 may be configured in such a way that their outputs (V_{D1} and V_{D2}) toggle at rising edges of V_{OC1} and V_{OC2}, respectively. The phase to PWM converter (406) may detect a time interval between the rising edges of V_{OC1} and V_{OC2}. Further, the phase to PWM converter (406) may convert the time interval between the rising edges of V_{OC1} and V_{OC2} into the on-time of the PWM output signal V_O, given as $VO = V_{D1} \oplus V_{D2}$.

[0061] Hence, the on-time (T_O) of V_O is proportional to the phase difference between V_{O3} and V_{S2}, illustrated in FIG. 4B. Hence, if T is the time period of the input signals (V_{S1} or V_{S2}), then the angle of rotation θ is given as,

$$\theta = 360^0 (T / T_O).$$

[0062] FIG. 5 illustrates an example flow chart for implementing a method (500) for detecting the angle of rotation of an object, in accordance with an embodiment of the present disclosure.

[0063] Referring to FIG. 5, the system may determine the angle of rotation of an object through a variable inductance method. The method (500) may include following steps for detecting the angle of rotation of an object.

[0064] At step 502, the method (500) may include detecting an inductance variation in stationary coils (202) based on an angle of rotation of an object. The sensor (102) may be configured with a shaft (204) attached to an object whose rotational angle to be measured. The shaft (204) may be surrounded by stationary coils (202), and a ferromagnetic strip (206) may be positioned on the shaft (204). As the shaft (204) rotates, the movement of the ferromagnetic strip (206) may induce inductance variations of the stationary coils (202).

[0065] At step 504, the method (500) may include converting the inductance variation in one or more stationary coils to voltage levels. The inductance to voltage converter (402) may be coupled to the stationary coils (202) of the sensor (104) and converts the detected inductance variations into corresponding voltage levels.

[0066] At step 506, the method (500) may include determining a phase difference between the voltage levels associated with inductance variation in the stationary coils (202). The phase difference determination circuit (404), such as formed using an instrumentation amplifier, may determine phase difference of voltage signals that is proportional to the angle of rotation of an object.

[0067] At step 508, the method (500) may include converting the phase difference between the voltage levels to a Pulse Width Modulation (PWM) to detect the angle of rotation of the object. The phase difference between voltage levels corresponding to the inductance variation and source voltage may be measured by the phase to PWM converter (406) of the interfacing module (106). The phase to PWM converter (406) may detect the phase difference between voltage levels corresponding to the inductance variation and source voltage as on-time of a PWM wave, where the on-time is directly proportional to the angle of rotation of an object.

EXPERIMENTS:

[0068] The proposed planar coil-based sensor system provides a full-circle (0◦-360◦) angle measurement using the variable reluctance principle. The system generates a pulse-width modulation (PWM) output, where the on-time is directly proportional to the angle of rotation. The system consists of a sensor and an interfacing module. The sensor incorporates two pairs of planar inductor coils wrapped around the shaft to form a hollow cylindrical structure, with a thin ferromagnetic material attached to a movable shaft. As the shaft rotates, the inductance of each coil changes, enabling precise angle measurement. Unlike conventional planar coil-based angle sensors, the proposed system offers a linear output, overcoming non-linearity issues. Additionally, the system is immune to misalignment of the shaft and variations in permeability of the ferromagnetic strip, addressing common challenges in angle sensing. The quasi-digital output eliminates need for microcontrollers, ADCs, or virtual instruments, simplifying system integration. The sensor ensures high-speed conversion with exceptional accuracy, making it ideal for industrial applications.

[0069] A prototype of the proposed sensor system was developed in the laboratory, and its performance was validated through various tests. Each coil pair is printed on a single PCB with dimensions 71 mm x 71 mm. These coils are arranged around the bobbin. A cylindrical bobbin, 40 mm in diameter and made of polylactic acid monomer (PLA), is used as the support structure for the planar coils. The movable part of the sensor, or shaft, is made from PLA and has an outer diameter of 29 mm. A ferromagnetic strip, made of μ-metal with an approximate relative permeability of 35,000 and a thickness of 4 mm, is attached to the shaft. The shaft is placed into the bobbin, ensuring there is an air gap of approximately 1 mm between the coils and the bobbin. An interfacing module was developed to measure the rotational angle of the shaft by monitoring the inductances L_{X1} to L_{X4} and to get the measured angle θ. The coils L_{X1} to L_{X4} are energized using an arbitrary function generator from NI ELVIS III+. Operational amplifiers OA_1 to OA_4 were implemented using the OP227P IC, while instrumentation amplifiers were realized using AD620 IC. Comparators OC_1 and OC_2 were built using the LM311 IC, and D-flip-flops DFF_1 and DFF_2 were implemented with the 74HCT74 IC. The XOR gate was realized using the 7486 IC. Various tests were conducted to assess the performance and efficacy of the proposed sensor system, as outlined below.

FUNCTIONALITY TESTS AND RESULTS:

[0070] 1) Linearity Test: To assess the linearity of the proposed sensor system, the shaft was rotated from $\theta = 0^\circ$ to 360° in increments of 10°. At each step, the pulse width (T_O) was recorded using an oscilloscope. The measured values of T_O and the corresponding measurement error at each angle θ are plotted in FIG. 6A. The worst-case linearity error observed during the measurement was 0.28%.

[0071] 2) Repeatability Test: The repeatability of the developed sensor system was evaluated by recording 150 consecutive readings. During the test, the shaft is fixed at angle $\theta = 110^\circ$ and the corresponding output was recorded. Performance metrics such as standard deviation, sensitivity, resolution, signal-to-noise ratio (SNR) and repeatability error were evaluated.

[0072] 3) Effect of Radial Misalignment of the shaft: To evaluate the system's insensitivity to shaft radial misalignment, the shaft was placed at four angular positions: $\theta = 0^\circ$, 90°, 180°, and 270°. At each position, the center of the shaft was displaced by -0.8 mm and +0.8 mm along the y axis and similarly along the x axis. The error in measurement for each case is evaluated. The maximum observed error is 0.2%, which is negligible.

[0073] 4) Effect due Presence of External Magnetic Material: The inductances of the sensor coils deviate from their expected values when a magnetic object is in proximity. To analyze this effect, a magnetic strip of length 80 mm and width 5 mm was placed at four different distances from the sensor coils: 1 mm, 2 mm, 3 mm, and 4 mm. At these distances, the error while measuring the angle observed was 0.35%, 0.31%, 0.2%, and 0.091%, respectively. The error at close distances can be further reduced by enclosing the sensor in a magnetic shield. The magnetic shield was realized using a iron metallic sheet with relative permeability of 5,000 and a thickness of 0.2 mm. The experiment was repeated with magnetic shield placed between the sensor and the external magnetic strip. The error was reduced to 0.14% at 1 mm distance.

[0074] 5) Effect due to external noise: To analyse the effect the noise on the interfacing circuit, a test was performed according to [30]. During this test, a white noise signal V_N is imparted along with the signals V_{S1} and V_{S2}. The magnitude of V_N is varied from 2 mV to 10 mV in steps of 1 mV. The worst-case error observed in the output was 0.47%. This result shows the immunity of the developed sensor system towards the external noise.

[0075] 6) Experiment at different rotational speeds of the shaft: To assess the performance of the proposed sensor system at various shaft rotation speeds, a DC servo motor is connected to the shaft. The speed of the motor is controlled by the motor controller. The shaft of the motor is coupled to an absolute encoder which serves as a reference to provide accurate speed and position control. The experiment was carried out at three different speeds: 200 rpm, 1000 rpm, and 1500 rpm. At each speed, the sensor output was compared to that of the reference encoder and the results at 1500 rpm are illustrated in FIG. 6B. The prototype sensor demonstrates excellent agreement with the reference sensor. The % error observed was 0.5%, 1.25%, and 1.64% at 200, 1000 and 1500 rpm, respectively. In another experiment, repeatability was tested by maintaining rotational speed of the shaft at 1000 rpm and measuring the output over ten repeated measurements. FIG. 6C illustrates three measurement cycles. The worst-case standard deviation observed was 0.014◦.

[0076] While the foregoing describes various embodiments of the disclosure, other and further embodiments of the invention may be devised without departing from the basic scope thereof. The scope of the disclosure is determined by the claims that follow. The disclosure is not limited to the described embodiments, versions, or examples, which are included to enable a person having ordinary skill in the art to make and use the disclosure when combined with information and knowledge available to the person having ordinary skill in the art.

ADVANTAGES OF THE PRESENT DISCLOSURE

[0077] The present disclosure provides a variable reluctance based system for detecting angle of rotation of an object which is inherently resistant to environmental factors thereby making the system suitable for applications in automotive, industrial, and aerospace sectors where such conditions prevail.

[0078] The present disclosure allows for cost-effective manufacturing, making the system an economical choice for large-scale industrial deployments.

[0079] The present disclosure provides a high-speed linear conversion of detected angle to a Pulse Width Modulation (PWM) output, thereby facilitating precise and real-time angle measurements.

We Claim:

1. A system (102) for detecting an angle of rotation of an object, the system (102) comprising:

a sensor (104) configured with a shaft (204) attached to the object,

wherein the shaft (204) is surrounded by one or more stationary coils (202); and

a ferromagnetic strip (206) positioned on the shaft (204), wherein an inductance of the one or more stationary coils (202) varies in accordance with a movement of the ferromagnetic strip (206); and

an interfacing module (106) electrically connected to the sensor (104), for converting inductance variations of the one or more stationary coils (202) to a Pulse Width Modulation (PWM) output to detect the angle of rotation of the object.

2. The system (102) as claimed in claim 1, wherein the one or more stationary coils (202) comprises at least two pairs of square shaped planar inductor coils positioned around the shaft (204) to form a hollow cylindrical structure.

3. The system (102) as claimed in claim 2, wherein the two pairs of square shaped planar inductor coils comprises a first pair of planar inductor coils that is positioned around the shaft (204) comprised in the system (102) and a second pair of planar inductor coils disposed surrounding a first surface of the first pair of planar inductor coils.

4. The system (102) as claimed in claim 2, wherein each of the first pair and second pair of planar inductor coils is bent over the shaft (204) to form the hollow cylindrical structure.

5. The system (102) as claimed in claim 3, wherein the first pair of planar inductor coils is arranged such that one planar coil of the first pair covers an angle of 0^0 to 180^0 and the second planar coil of the first pair covers an angle of 180◦ to 360◦ of the hollow cylindrical structure.

6. The system (102) as claimed in claim 3, wherein the second pair of planar inductor coils is arranged such that one planar coil of the second pair covers an

angle of 90^0 to 270^0 and the second planar coil of the second pair covers an angle of 270° to 90° of the hollow cylindrical structure.

7. The system (102) as claimed in claim 1, wherein the interfacing module (106) comprises an inductance to voltage converter (402) and configured to convert the inductance variation in the one or more stationary coils (202) to corresponding voltage levels.

8. The system (102) as claimed in claim 1, wherein the interfacing module (106) comprises a phase difference determination circuit (404) and configured to determine phase of the voltage levels.

9. The system (102) as claimed in claim 1, wherein the interfacing module (106) comprises a phase to PWM converter (406) and configured to convert phase difference between the voltage levels associated with the inductance variation in the one or more stationary coils to the PWM output.

10. A method (500) for detecting an angle of rotation of an object, the method (500) comprising:

detecting, by a system (102), an inductance variation in one or more stationary coils (202) based on an angle of rotation of an object;

converting, by the system (102), the inductance variation in the one or more stationary coils (202) to voltage levels;

determining, by the system (102), a phase difference between the voltage levels associated with inductance variation in the one or more stationary coils (202);

converting, by the system (102), the phase difference between the voltage levels to a Pulse Width Modulation (PWM) output to detect the angle of rotation of the object.

For Indian Institute of Technology, Palakkad and
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ABSTRACT

A PLANAR-COIL VARIABLE RELUCTANCE BASED ANGLE SENSOR SYSTEM WITH PWM OUTPUT

The present disclosure provides a system (102) and a method (500) for precise rotational angle measurement of an object. The system (102) includes variable reluctance principle to achieve full-circle (0°–360°) angle measurement. The system (102) includes a sensor (104) and an interfacing module (106). The sensor (104) includes two pairs of planar inductor coils wrapped around a shaft (204), forming a cylindrical structure, with a thin ferromagnetic strip (206) attached to the shaft (204). As the shaft (204) rotates, inductance of each coil varies, enabling accurate angle measurement. The interfacing module (106) of the system (102) generates a Pulse-Width Modulation (PWM) output based on inductance variation of the planar inductor coils, where on-time of the PWM signal is directly proportional to the angle of rotation of an object. The system provides high-speed linear conversion of the detected angle to the PWM output.

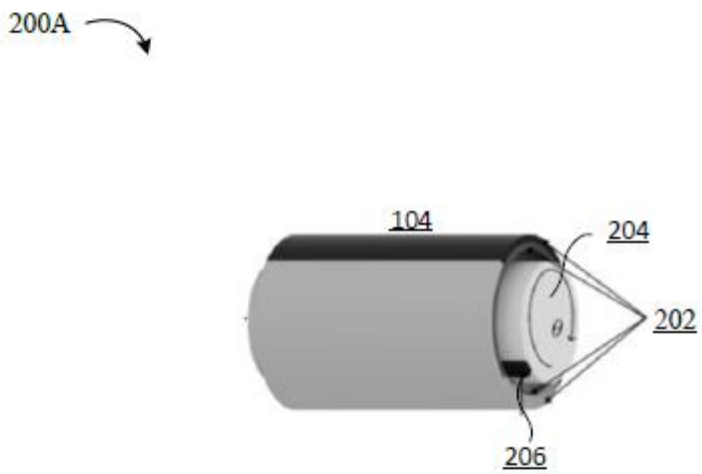

FIG. 2A